

Chapter 15: Fenchel-Rockafellar Duality

Convex Analysis and Monotone Operator Theory

Based on: Convex Analysis and Monotone Operator Theory in Hilbert Spaces (2e, 2017)

2024

Chapter 15 Overview

1 15.1: The Attouch-Brézis Theorem

- Exact formula for sum of conjugate functions
- Infimal convolution properties

2 15.2: Fenchel Duality

- Primal-dual optimization problems
- Duality gap and optimality conditions

3 15.3: Fenchel-Rockafellar Duality

- Duality with linear operators
- Zero duality gap conditions

4 15.4: A Conjugation Result

- Conjugate of composite functions

5 15.5: Applications

- Von Neumann's minimax theorem
- Convex cone and operator results

Key Concepts

Fenchel Conjugate: For $f : \mathcal{H} \rightarrow [-\infty, +\infty]$,

$$f^*(u) = \sup_{x \in \mathcal{H}} (\langle u, x \rangle - f(x))$$

Infimal Convolution: For $f, g \in \Gamma_0(\mathcal{H})$,

$$(f \square g)(x) = \inf_{y \in \mathcal{H}} [f(y) + g(x - y)]$$

Primal-Dual Problem:

- **Primal:** $\min_x [f(x) + g(x)]$
- **Dual:** $\min_u [f^*(-u) + g^*(u)]$

The Attouch-Brézis Theorem

Theorem 15.3 (Attouch-Brézis)

Let f and g be functions in $\Gamma_0(\mathcal{H})$ such that

$$0 \in \text{sri}(\text{dom } f - \text{dom } g)$$

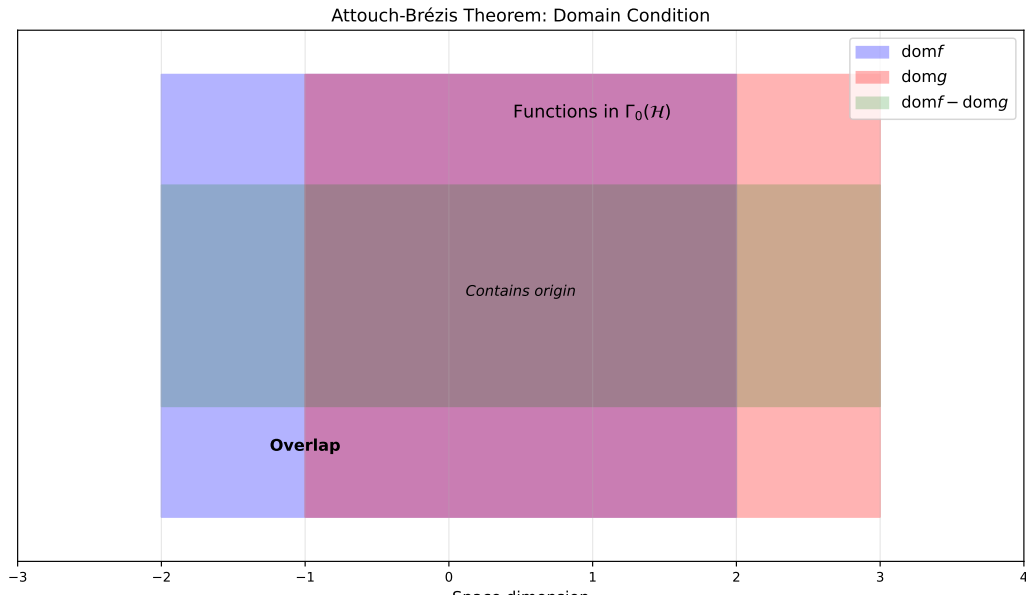
where sri denotes the **strong relative interior**.

Then:

$$(f + g)^* = f^* \square g^*$$

Key Insight: The conjugate of a sum equals the infimal convolution of conjugates under a constraint on domain difference.

Illustration: Attouch-Brézis



Proposition 15.5: Sufficient Conditions

Proposition 15.5

Let $f, g \in \Gamma_0(\mathcal{H})$ with $\text{dom} f \cap \text{dom} g \neq \emptyset$.

Any one of these implies $0 \in \text{sri}(\text{dom} f - \text{dom} g)$:

- (i) $\text{cone}(\text{dom} f - \text{dom} g) = \text{span}(\text{dom} f - \text{dom} g)$
- (ii) $0 \in \text{core}(\text{dom} f - \text{dom} g)$
- (iii) $0 \in \text{int}(\text{dom} f - \text{dom} g)$
- (iv) $\text{cont} f \cap \text{dom} g \neq \emptyset$
- (v) $\mathcal{H}$ is finite-dimensional and $\text{ri}(\text{dom} f) \cap \text{ri}(\text{dom} g) \neq \emptyset$

Example: Attouch-Brézis Application

Example: Let $\mathcal{H} = \mathbb{R}$, $f(x) = |x|$, $g(x) = x^2 + 1$.

Conjugates:

- $f^*(u) = \begin{cases} 0 & \text{if } |u| \leq 1 \\ +\infty & \text{otherwise} \end{cases}$
- $g^*(u) = \frac{u^2}{4} - 1$

Computation of $(f + g)^*$:

```
# Python example
import numpy as np
f = lambda x: np.abs(x)
g = lambda x: x**2 + 1
combined = lambda x: f(x) + g(x)

# The conjugate property holds when domains satisfy the condition
#  $(f + g)^*(u) = \inf_v [f^*(v) + g^*(u - v)]$ 
```

The Attouch-Brézis theorem ensures exact conjugacy formula.

Remark 15.4: Tightness of Conditions

Key Remarks

- ① If $\mathcal{H} = \mathbb{R}^2$ and $\text{dom} f = \mathbb{R} \times \{0\}$, $\text{dom} g = \{0\} \times \mathbb{R}$, then

$$(f + g)^* = \iota_{\{0\}} \quad \text{but} \quad f^* \square g^* \neq \iota_{\{0\}}$$

- ② Theorem 15.3 can fail if strong relative interior is replaced by relative interior.
③ Lower semicontinuity of f and g is necessary.

These examples show conditions in Theorem 15.3 are tight and cannot be weakened.

Fenchel Duality: Basic Framework

Definition 15.10

For proper functions $f, g : \mathcal{H} \rightarrow [-\infty, +\infty]$:

Primal Problem:

$$\min_{x \in \mathcal{H}} [f(x) + g(x)]$$

Dual Problem:

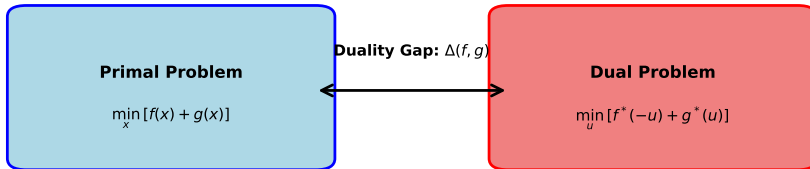
$$\min_{u \in \mathcal{H}} [f^*(-u) + g^*(u)]$$

Duality Gap:

$$\Delta(f, g) = \begin{cases} 0 & \text{if } \mu = -\mu^* \in [-\infty, +\infty] \\ \mu + \mu^* & \text{otherwise} \end{cases}$$

where $\mu = \inf(f + g)(\mathcal{H})$ and $\mu^* = \inf(f^* + g^*)(\mathcal{H})$.

Fenchel Duality Gap Analysis



Fenchel-Rockafellar Duality

If $0 \in \text{sri}(\text{dom} f - \text{dom} g)$, then:

$$\min_x [f(x) + g(x)] = -\min_u [f^*(-L^* u) + g^*(u)]$$

Zero duality gap: $\Delta(f, g, L) = 0$

Proposition 15.9: Weak Duality

Proposition 15.9

For proper functions $f, g : \mathcal{H} \rightarrow [-\infty, +\infty]$:

$$(\forall x \in \mathcal{H})(\forall u \in \mathcal{H}) \quad f(x) + g(x) \geq -f^*(-u) - g^*(u)$$

Hence:

$$\inf(f + g)(\mathcal{H}) \geq -\inf(f^* + g^*)(\mathcal{H})$$

Proof Sketch: Using Fenchel-Young inequality:

$$\langle u, x \rangle \leq f(x) + f^*(u) \quad \text{and} \quad \langle u, x \rangle \leq g(x) + g^*(u)$$

Proposition 15.12: Duality Properties

Proposition 15.12

Let f, g be proper, and set

$$\mu = \inf(f + g)(\mathcal{H}), \quad \mu^* = \inf(f^* + g^*)(\mathcal{H})$$

Then:

- (i) $\mu \geq -\mu^*$ (weak duality)
- (ii) $\Delta(f, g) \in [0, +\infty]$ (gap is non-negative)
- (iii) $\mu = -\mu^* \Leftrightarrow \Delta(f, g) = 0$ (strong duality)

Strong duality ($\Delta(f, g) = 0$) means optimal values are negatives of each other.

Corollaries: Special Cases

Corollary 15.14

If $f \in \Gamma_0(\mathcal{H})$ and K is a closed convex cone with $0 \in \text{sri}(\text{dom} f - K)$:

$$\inf f(K) = -\min f^*(K^\perp)$$

Corollary 15.15

If $f, g \in \Gamma_0(\mathcal{H})$ with $0 \in \text{sri}(\text{dom} f - \text{dom} g)$, $f + g \geq 0$, and $g^* = g \circ L$ for $L \in \mathcal{B}(\mathcal{H})$:

$$\exists v \in \mathcal{H} : f^*(-v) + g(Lv) \leq 0$$

Corollary 15.17

For $f \in \Gamma_0(\mathcal{H})$ and $q(x) = \frac{1}{2}\|x\|^2$ with $f + q \geq 0$:

$$\exists w \in \mathcal{H} : (\forall x) f(x) + q(x) \geq q(x - w)$$

Fenchel-Rockafellar Duality: Setup

Definition 15.19

For $f : \mathcal{H} \rightarrow [-\infty, +\infty]$, $g : \mathcal{K} \rightarrow [-\infty, +\infty]$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$:

Primal Problem:

$$\min_{x \in \mathcal{H}} [f(x) + g(Lx)]$$

Dual Problem:

$$\min_{v \in \mathcal{K}} [f^*(-L^*v) + g^*(v)]$$

Duality Gap:

$$\Delta(f, g, L) = \begin{cases} 0 & \text{if } \mu = -\mu^* \\ \mu + \mu^* & \text{otherwise} \end{cases}$$

Proposition 15.18: Weak Duality with Operators

Proposition 15.18

Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ be proper, $g : \mathcal{K} \rightarrow [-\infty, +\infty]$ be proper, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$.
Then:

$$(\forall x \in \mathcal{H})(\forall v \in \mathcal{K}) \quad f(x) + g(Lx) \geq -f^*(-L^*v) - g^*(v)$$

and

$$\inf(f + g \circ L)(\mathcal{H}) \geq -\inf(f^* \circ (-L^*) + g^*)(\mathcal{K})$$

Key: The operator L and its adjoint L^* couple the primal and dual spaces.

Theorem 15.23: Strong Duality

Theorem 15.23

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ be such that

$$0 \in \text{sri}(\text{dom} g - L(\text{dom} f))$$

Then:

$$\inf(f + g \circ L)(\mathcal{H}) = -\min(f^* \circ (-L^*) + g^*)(\mathcal{K})$$

Proof Strategy:

- 1 Use projection onto relevant subspaces
- 2 Apply Corollary 15.14 to conjugate functions
- 3 Exploit the condition on domain difference

Proposition 15.22-15.24: Sufficient Conditions

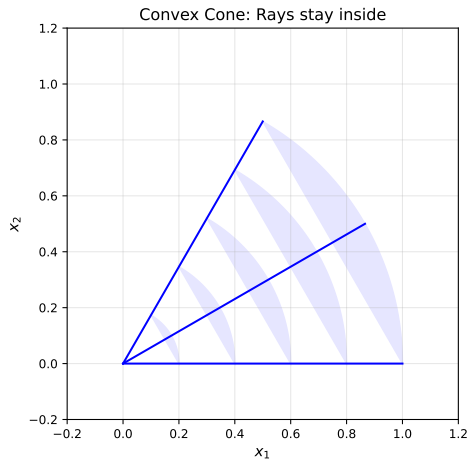
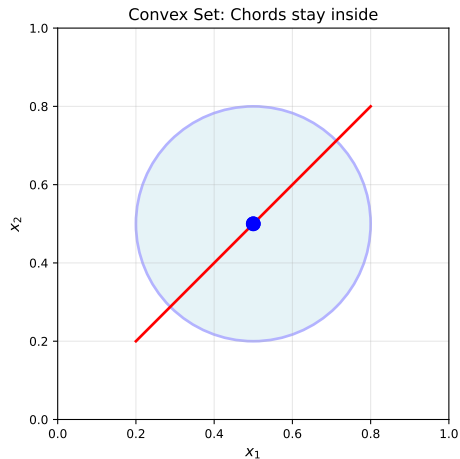
Proposition 15.24

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ with $\text{dom} g \cap L(\text{dom} f) \neq \emptyset$.

Any of these implies $0 \in \text{sri}(\text{dom} g - L(\text{dom} f))$:

- (i) $\text{cone}(\text{dom} g - L(\text{dom} f)) = \text{span}(\text{dom} g - L(\text{dom} f))$
- (ii) $\text{dom} g - L(\text{dom} f)$ is closed and linear
- (iii) $\text{dom} f, \text{dom} g$ linear and $\text{dom} g + L(\text{dom} f)$ closed
- (iv) g cone, $\text{dom} g - \text{cone} L(\text{dom} f)$ closed and linear
- (v) $0 \in \text{core}(\text{dom} g - L(\text{dom} f))$
- (vi) $0 \in \text{int}(\text{dom} g - L(\text{dom} f))$
- (vii) $\text{cont} g \cap L(\text{dom} f) \neq \emptyset$
- (viii) $\mathcal{K}$ finite-dimensional and $\text{ri}(\text{dom} g) \cap \text{ri} L(\text{dom} f) \neq \emptyset$

Convex Sets and Conjugate Functions



Convex sets and cones are fundamental to understanding domain conditions in duality.

Theorem 15.27: Conjugate of Composite Function

Theorem 15.27

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$.

Suppose one of:

- (i) $0 \in \text{sri}(\text{dom} g - L(\text{dom} f))$
- (ii) $\mathcal{K}$ finite-dim., g polyhedral, $\text{dom} g \cap L(\text{dom} f) \neq \emptyset$
- (iii) $\mathcal{H}, \mathcal{K}$ finite-dim., f, g polyhedral, $\text{dom} g \cap L(\text{dom} f) \neq \emptyset$

Then:

$$(f + g \circ L)^*(u) = \min_{v \in \mathcal{K}} [f^*(u - L^*v) + g^*(v)]$$

Key Formula: The conjugate of a composite function is a constrained minimization.

Corollary 15.28: Simplified Case

Corollary 15.28

Let $g \in \Gamma_0(\mathcal{K})$ and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$.

Suppose one of:

- (i) $0 \in \text{sri}(\text{dom} g - \text{ran} L)$
- (ii) $\mathcal{K}$ finite-dim., g polyhedral, $\text{dom} g \cap \text{ran} L \neq \emptyset$

Then:

$$(g \circ L)^*(u) = L^* \circ g^*(u) = \min_{v \in \mathcal{K}, L^*v=u} g^*(v)$$

Application 1: Von Neumann's Minimax Theorem

Corollary 15.30 (Von Neumann)

Let C be a nonempty bounded closed convex subset of $\mathcal{H}$, D a nonempty bounded closed convex subset of $\mathcal{K}$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$.

Then:

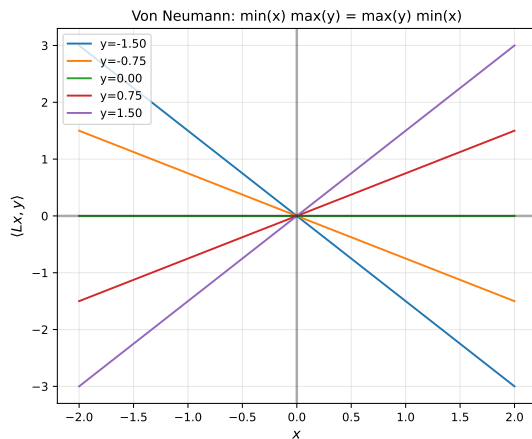
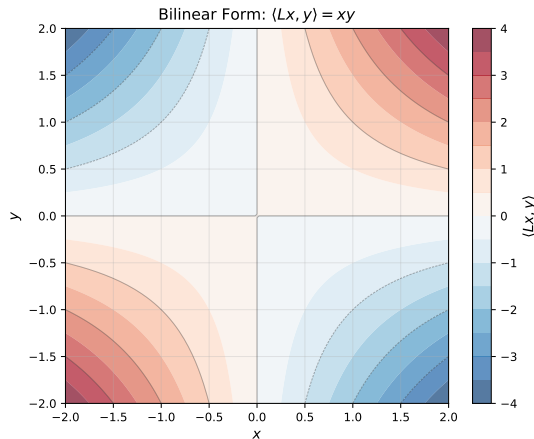
$$\min_{x \in C} \max_{v \in D} \langle Lx, v \rangle = \max_{v \in D} \min_{x \in C} \langle Lx, v \rangle$$

Proof: Set $f = \iota_C$, $g = \iota_D$, then:

$$\min_x [f(x) + g(Lx)] = \min_x \iota_C(x) + \iota_D(Lx)$$

Apply Fenchel-Rockafellar duality with $f^* = \sigma_C$, $g^* = \sigma_D$ (support functions).

Application 2: Minimax Visualization



Von Neumann's theorem: order of min and max can be exchanged under convexity conditions.

Application 3: Conjugate Cones and Polars

Corollary 15.31

Let C, D be nonempty closed convex cones, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$.

Then:

- (i) $(C \cap L^{-1}(D))^{\circ} = C^{\circ} + L^{*}(D^{\circ})$
- (ii) If $D - L(C)$ is closed linear subspace:

$$(C \cap L^{-1}(D))^{\circ} = C^{\circ} + L^{*}(D^{\circ}) \text{ is nonempty closed convex cone}$$

These results characterize closure of sums and compositions of cones.

Application 4: Operator Range Closedness

Corollary 15.34

Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. Then $\text{ran} L$ is closed iff $\text{ran} L^*$ is closed.

Corollary 15.35

Let U, V be closed linear subspaces of $\mathcal{H}$. Then $U + V$ is closed iff $U^\perp + V^\perp$ is closed.

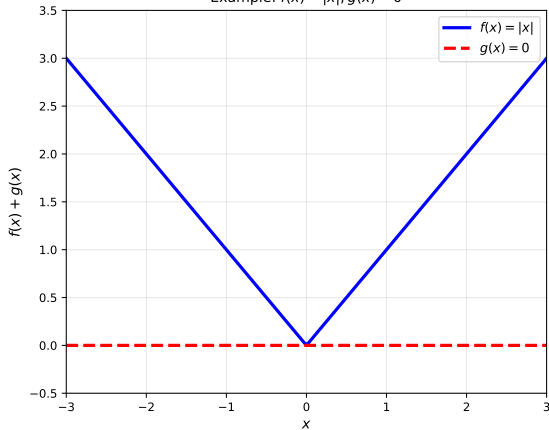
Corollary 15.36

Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ with closed range and V a closed subspace. Then $L(V)$ is closed iff $V + \ker L$ is closed.

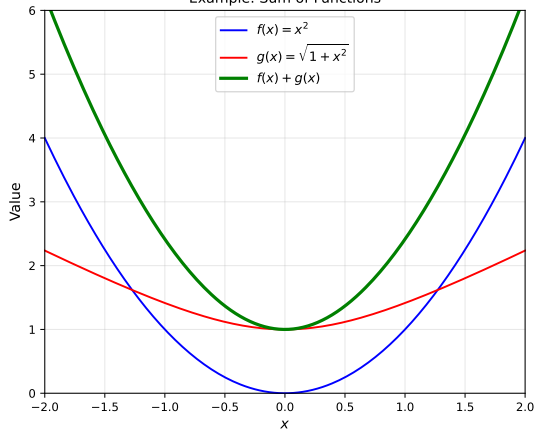
Impact: These characterizations are essential in functional analysis.

Key Properties Summary

Example: $f(x) = |x|, g(x) = 0$



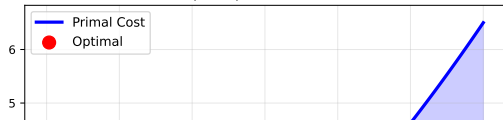
Example: Sum of Functions



Example: Function and Dual



Example: Optimization Problem



Example 1: Quadratic + Absolute Value

Problem: $\min_x \left[\frac{1}{2} \|x\|_2^2 + |x_1| + |x_2| \right]$

```
import numpy as np
from scipy.optimize import minimize

# Primal problem
def objective(x):
    return 0.5*np.sum(x**2) + np.sum(np.abs(x))

x0 = np.zeros(2)
result = minimize(objective, x0, method='BFGS')
print(f"Optimal x: {result.x}")
print(f"Optimal value: {result.fun}")

# Dual conjugates:
# f(x) = 0.5||x||^2, f*(u) = 0.5||u||^2
# g(x) = |x|, g*(u) = 0 if ||u||_inf <= 1, else inf
```

The strong duality condition is satisfied, so primal and dual optimal values are equal.

Example 2: Linear Constraint

Problem: $\min_x f(x)$ subject to $Lx = y$ (encoded in conjugate)

```
import numpy as np

# Primal: min [f(x) + g(Lx)] where g = iota_{y}
# Dual: min [f*(-L*v) + g*(v)]

# Example: L is linear map, y is target
L = np.array([[1, 1], [1, -1]])
y = np.array([1, 0])

def primal_objective(x):
    residual = L @ x - y
    return np.linalg.norm(residual)**2

# Dual via Fenchel-Rockafellar:
# min_v ||L^T v||^2 / 4 + ||y - v||^2
def dual_objective(v):
    return np.linalg.norm(L.T @ v)**2 / 4 + \
           np.linalg.norm(y - v)**2
```

Example 3: Composite Functions

Computing Conjugate of $f(x) + g(Lx)$

```
import numpy as np

# Given: f, g, L
# Compute: (f + g @ L)*(u)

def conjugate_of_composite(u, f_star, g_star, L_star):
    """
    Theorem 15.27:
    (f + g @ L)*(u) = min_v [f*(u - L*v) + g*(v)]
    """
    def objective(v):
        return f_star(u - L_star @ v) + g_star(v)

    from scipy.optimize import minimize
    result = minimize(objective, np.zeros_like(u))
    return result.fun

# Example: f(x) = ||x||^2/2, g(x) = |x|, L = I
f_star = lambda u: np.linalg.norm(u)**2 / 2
L_star = np.eye(2)
```

Chapter 15 Summary

Major Results:

① **Attouch-Brézis:** $(f + g)^* = f^* \square g^*$ under domain conditions

② **Fenchel Duality:** Primal-dual framework with duality gap

$$\inf(f + g) = -\inf(f^* + g^*) \text{ if } \Delta(f, g) = 0$$

③ **Fenchel-Rockafellar:** Extension with linear operators

$$\inf(f + g \circ L) = -\min(f^* \circ (-L^*) + g^*)$$

④ **Conjugate Formula:** For composite functions

$$(f + g \circ L)^*(u) = \min_v [f^*(u - L^*v) + g^*(v)]$$

⑤ **Applications:**

- Von Neumann's minimax theorem
- Characterization of closed operator ranges
- Properties of convex cones and subspaces

Key Conditions

Sufficient Conditions for Strong Duality:

Most results require one of:

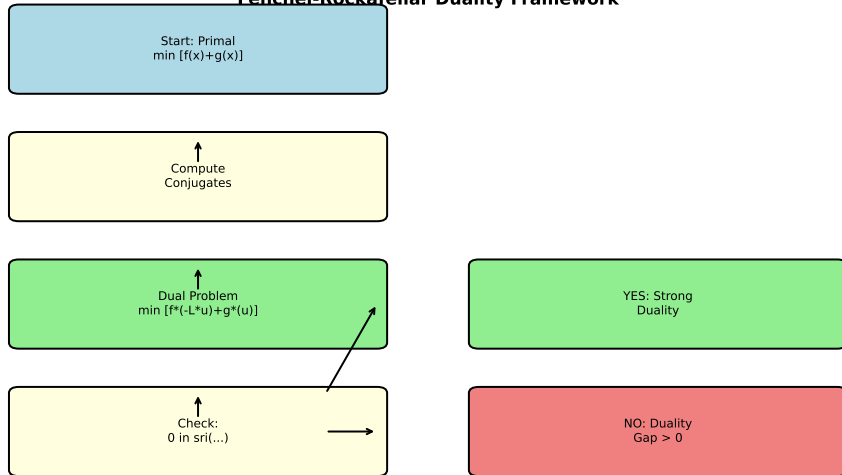
- ① $0 \in \text{sri}(\text{dom} f - \text{dom} g)$ (strong relative interior)
- ② $0 \in \text{int}(\text{dom} f - \text{dom} g)$ (interior)
- ③ $0 \in \text{core}(\text{dom} f - \text{dom} g)$ (core)
- ④ Relative interior conditions in finite dimensions
- ⑤ Explicit finiteness and polyhedrality assumptions

Why These Conditions?

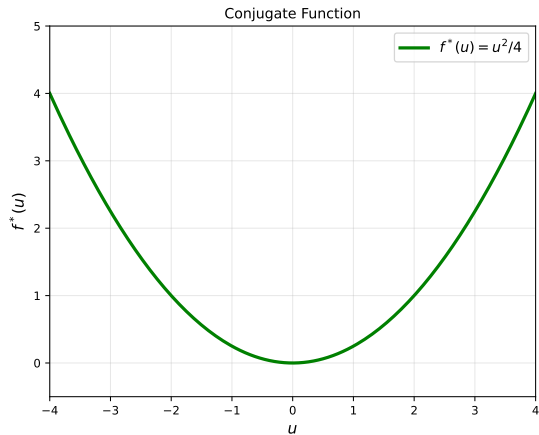
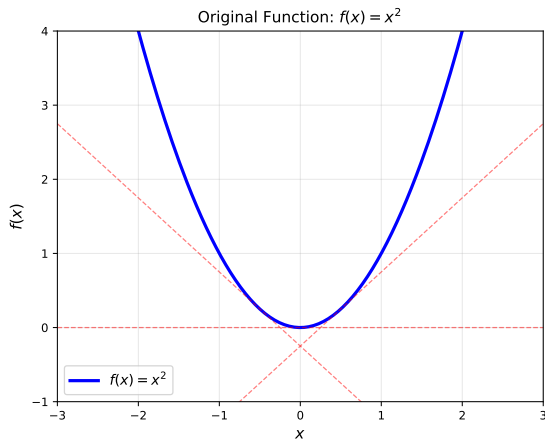
- Ensure properness and lower semicontinuity
- Guarantee exact conjugacy formulas
- Avoid pathological examples with duality gap

Duality Framework Visualization

Fenchel-Rockafellar Duality Framework



Fenchel Conjugate Visualization



Computational Perspective

Practical Algorithm:

```
# Step 1: Identify primal problem structure
# min_x [f(x) + g(Lx)]

# Step 2: Check domain condition
# 0 in sri(dom g - L(dom f))?

# Step 3: Compute conjugate functions
# f*(u) = sup_x [<u,x> - f(x)]
# g*(v) = sup_y [<v,y> - g(y)]

# Step 4: Formulate dual problem
# min_v [f*(-L*v) + g*(v)]

# Step 5: Solve dual (often easier)
# Use convex optimization methods

# Step 6: Recover primal solution
# From strong duality: optimal values match
# Can extract x from v via optimality conditions
```

This approach is fundamental in optimization algorithms.

Further Reading

- **Boyd & Vandenberghe** (2004): Convex Optimization
 - Standard reference for convex duality
- **Rockafellar** (1970): Convex Analysis
 - Foundational monograph on Fenchel duality
- **Nesterov** (2004): Introductory Lectures on Convex Optimization
 - Algorithmic perspective on duality
- **Bauschke & Combettes** (2011): Convex Analysis and Monotone Operator Theory
 - Comprehensive treatment (this book)

Thank you!

Chapter 15: Fenchel-Rockafellar Duality

Questions?